

Assessment model continuum and data limited methods

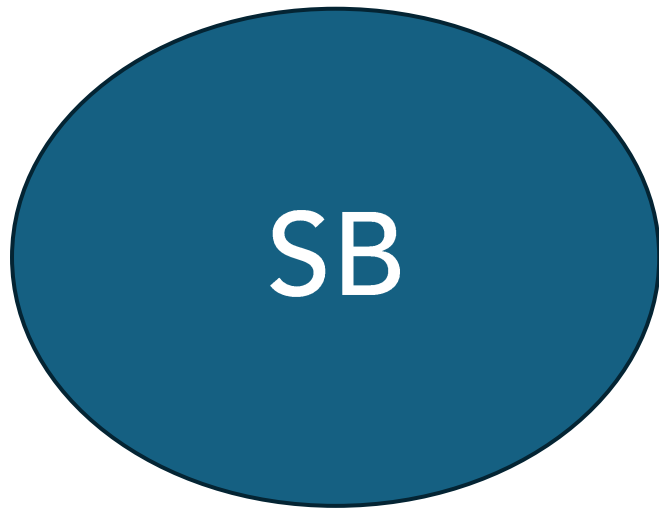
MAR 580 Fall 2025

21 & 23 October

Jason Cope

Stock assessments: 1 lump or many?

1 lump



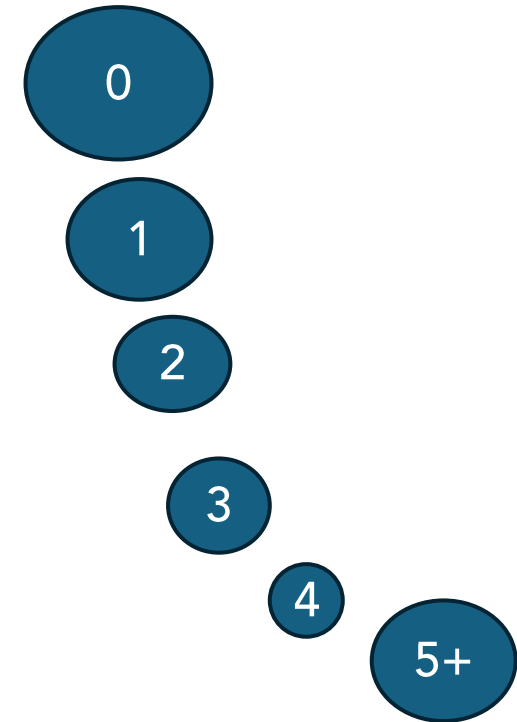
Ex: Surplus
Production

2 lumps



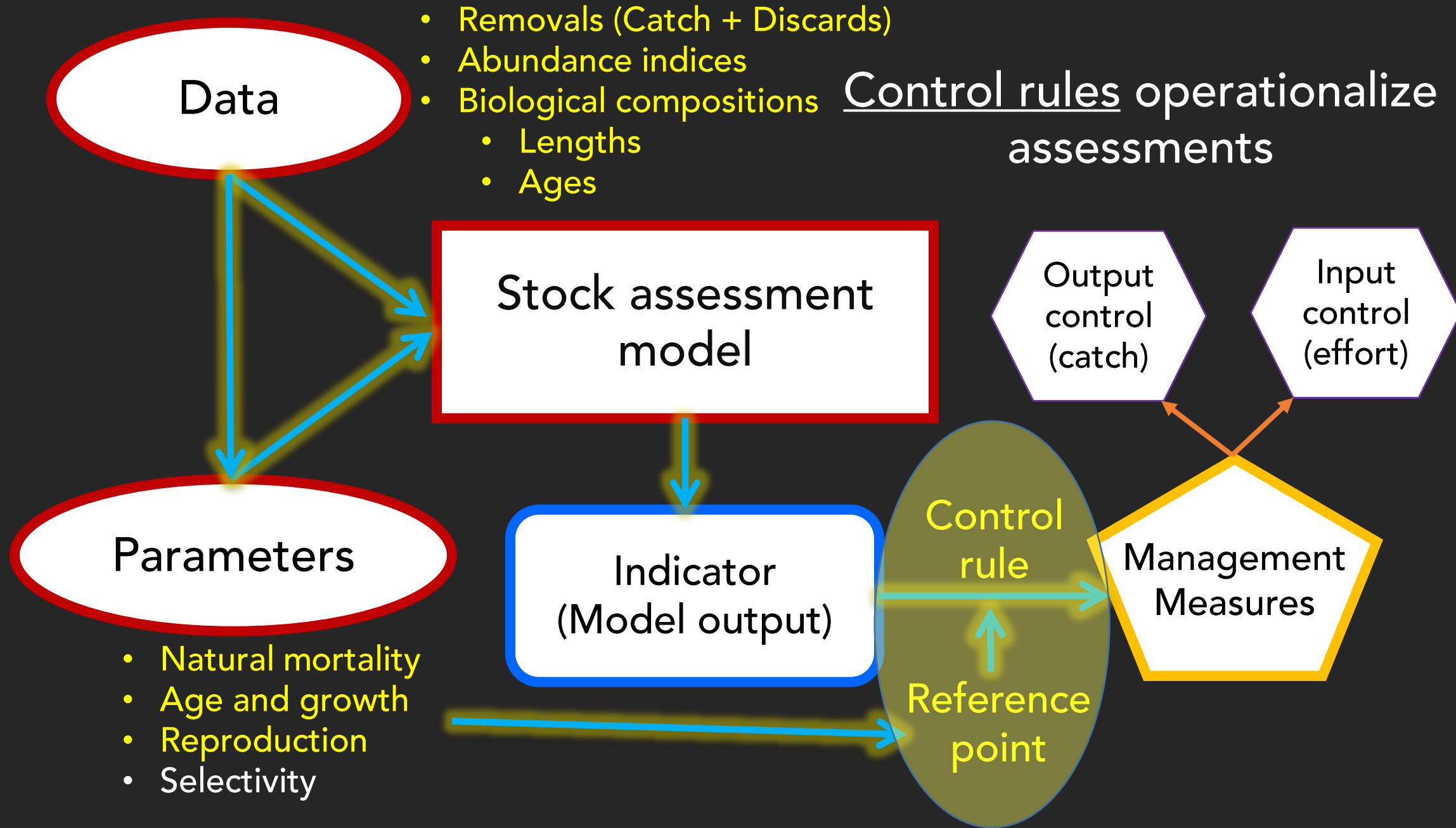
Ex: Delay
Difference

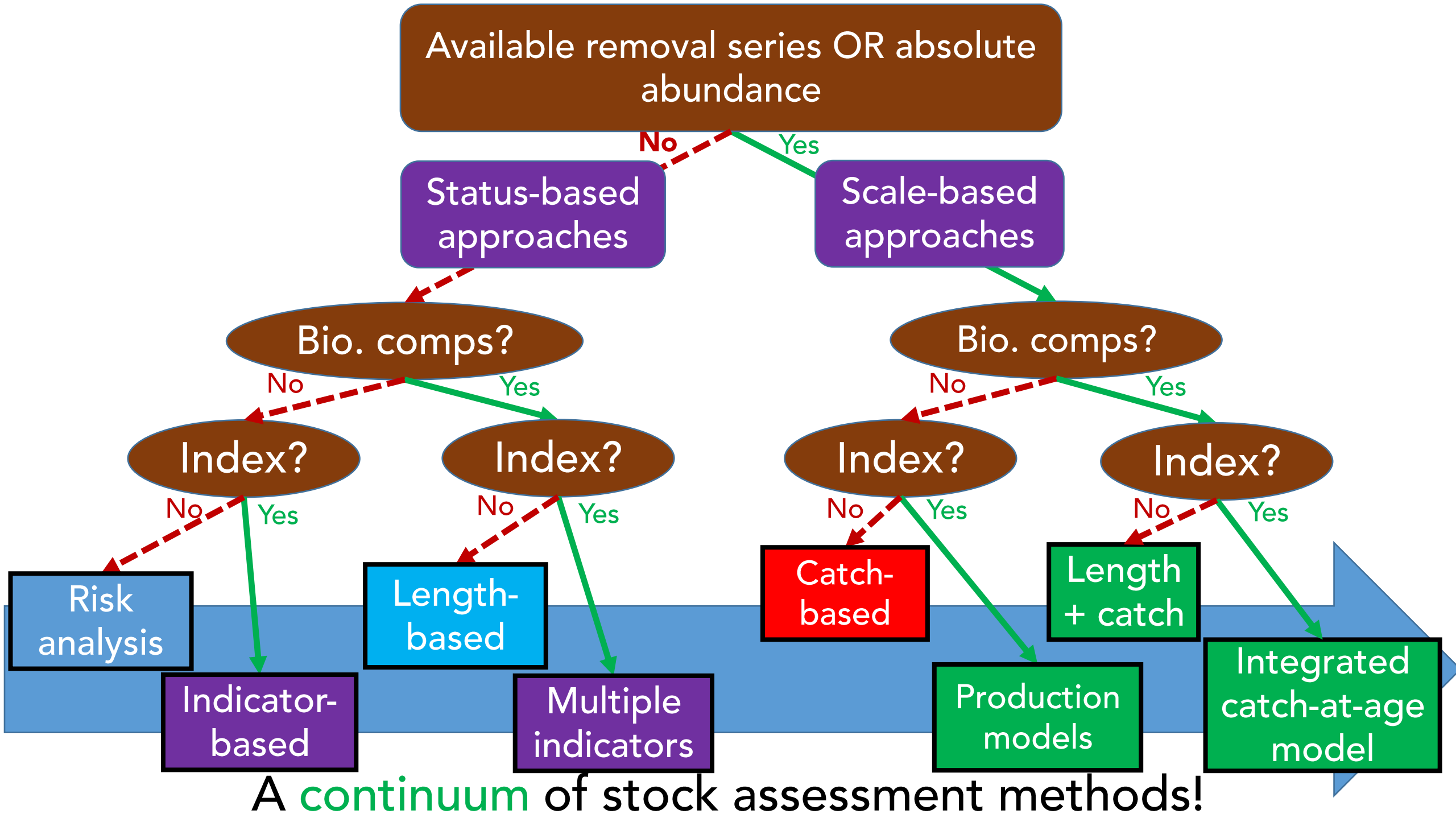
Many lumps



Ex: Age
structured

Basics of stock assessment





SAC Tool

Welcome to the Stock Assessment Continuum Tool, powered by Stock Synthesis (version 3.30.22.1)

(Formerly known as the Stock Synthesis Data-Limited (SS-DL) tool), this tool uses the **Stock Synthesis** framework to implement a **variety of types** of models.

Any suggested changes or requests? Please submit an issue with the recommendation [here](#)

Access the latest version of the Stock Synthesis manual [here](#)

Watch a talk on how to use the SAC tool and the concept of the stock assessment continuum [here](#)

Upload bookmarked inputs

Upload a saved input rds file:

Browse...

File names should only include letters and numbers, otherwise an upload error may occur. If you see 'Error: Invalid state id', rename the file and try again.

Choose data file

Catch time series

Browse...

Length composition

Browse...

Age composition

Browse...

Abundance index

Browse...

Clear data files

Catches

Length

Ages

Index

☐ Use existing model files?

Data and Parameters

Model output

Model efficiency

Jitter outputs

Likelihood profile

Retrospectives

Sensitivity Plots

Ensemble models

Life history

Selectivity

SAC Tool

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The screenshot shows the SAC Tool interface. At the top, there's a navigation bar with tabs: 'Data and Parameters', 'Model output', 'Model efficiency', 'Jitter outputs', 'Likelihood profile', 'Retrospectives', 'Sensitivity Plots', and 'Ensemble models'. Below this, the 'Data and Parameters' tab is active, showing a 'Life history' section. On the left, there are three main sections: 'Upload bookmarked inputs' (with a 'Browse...' button), 'Choose data file' (with four 'Browse...' buttons for 'Catch time series', 'Length composition', 'Age composition', and 'Abundance index'), and 'Clear data files' (with buttons for 'Catches', 'Length', 'Ages', and 'Index'). A checkbox at the bottom says 'Use existing model files?'. Three arrows point from the interface to a list of actions: a purple arrow from the 'Upload bookmarked inputs' section to 'Load inputs', a green arrow from the 'Choose data file' section to 'Load data' and 'Configures SS3 depending on data availability', and a blue arrow from the 'Clear data files' section to 'Clear data'.

Upload bookmarked inputs

Upload a saved input rds file:

Browse...

File names should only include letters and numbers, otherwise an upload error may occur. If you see 'Error: Invalid state id', rename the file and try again.

Choose data file

Catch time series

Browse...

Length composition

Browse...

Age composition

Browse...

Abundance index

Browse...

Clear data files

Catches Length Ages Index

☐ Use existing model files?

Life history

Selectivity

- Load inputs
- Load data
- Configures SS3 depending on data availability
 - SSS
 - SS-LO
 - SS-LAO
 - SS-CL
 - SS-ASPM
 - SS...
- Clear data

SAC Tool

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Upload bookmarked inputs

Upload a saved input rds file:

Browse...

File names should only include letters and numbers, otherwise an upload error may occur. If you see 'Error: Invalid state id', rename the file and try again.

Choose data file

Catch time series

Browse...

Length composition

Browse...

Age composition

Browse...

Abundance index

Browse...

Clear data files

Catches

Length

Ages

Index

☒ Use existing model files?

Data and Parameters

Model output

Model efficiency

Jitter outputs

Likelihood profile

Retrospectives

Sensitivity Plots

Ensemble models

Life history

Selectivity

Using existing model files

- Allows custom model specification
- Still access SAC Tool

SAC Tool: Model input

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Choose data file

Catch time series

Browse... Catches.csv

Upload complete

Length composition

Browse... Lengths.csv

Upload complete

Age composition

Browse...

Abundance index

Browse...

Clear data files

Catches

Length

Ages

Index

☐ Use existing model files?

Do not use this option with catch only models.

Make sure the model run is in the Scenarios folder. Put that folder name in the Scenario name input and run the model.

Using an existing model allows you to do run complex and custom model runs outside the options of the SS-DL tool, but still use some of the quick features.

Examples are applying jitters or any of the additional SS3 options that do not modify the data or control files.

To do sensitivity runs for pre-existing models, make a copy of the folder and re-name it, then make desired changes in the data and/or control files.

Data-weighting

Data weighting balances information content of biological data with model structure

Data and Parameters

Model output

Model efficiency

Jitter exploration

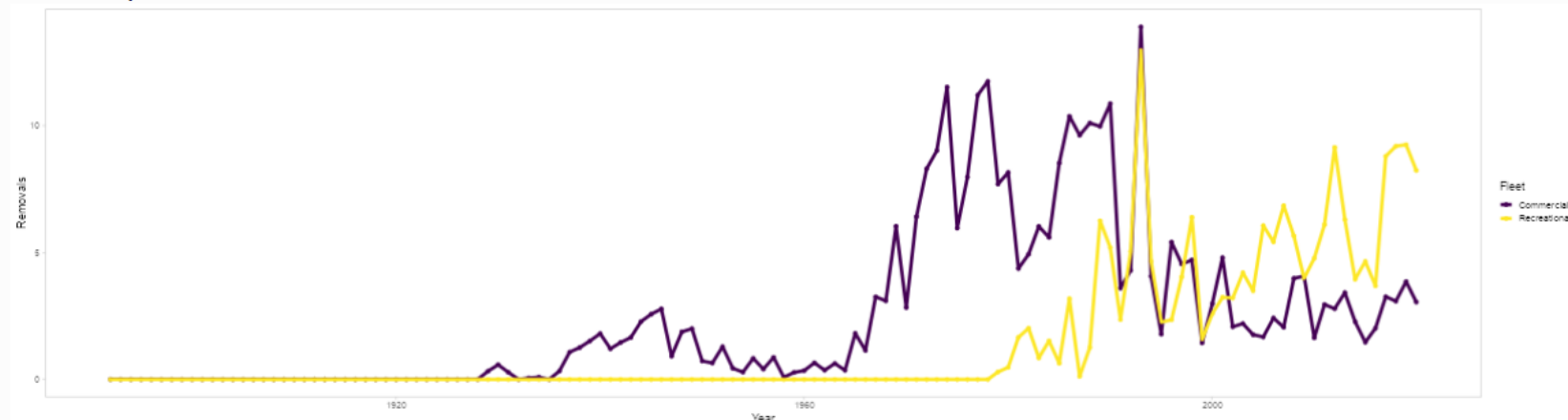
Likelihood profile

Retrospectives

Sensitivity Plots

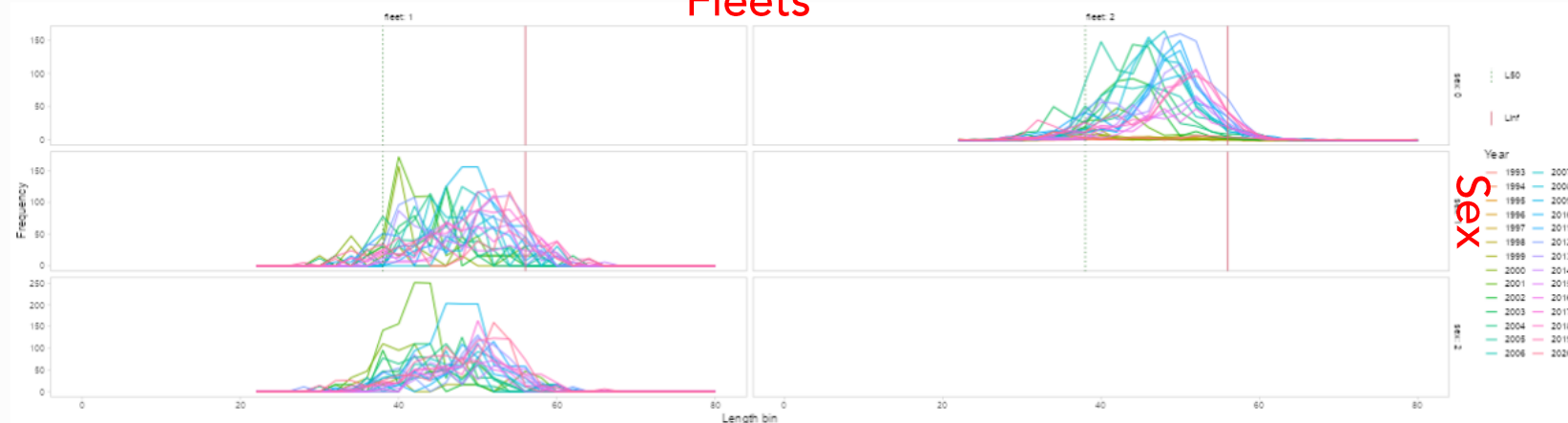
Ensemble models

Removal history



Length compositions. Sex categories: unknown sex = 0, females = 1 and males = 2.

Fleets



Life history



SAC Tool: Life history

Life history inputs

Female

Natural mortality

Asymptotic size (L_{∞})

Growth coefficient k

Age at length 0 (t_0)

CV at length (young then old)

Length at 50% maturity

Length at 95% maturity

Length-weight relationship $W=aL^b$. Weight is in kg and length in cm.

a in $W=aL^b$

b in $W=aL^b$

Length-fecundity relationship $F=aL^b$. Fecundity is measured in number of eggs or pups. Set equal to the length-weight relationship to get fecundity equivalent to spawning biomass.

a in $F=aL^b$

b in $F=aL^b$

Male

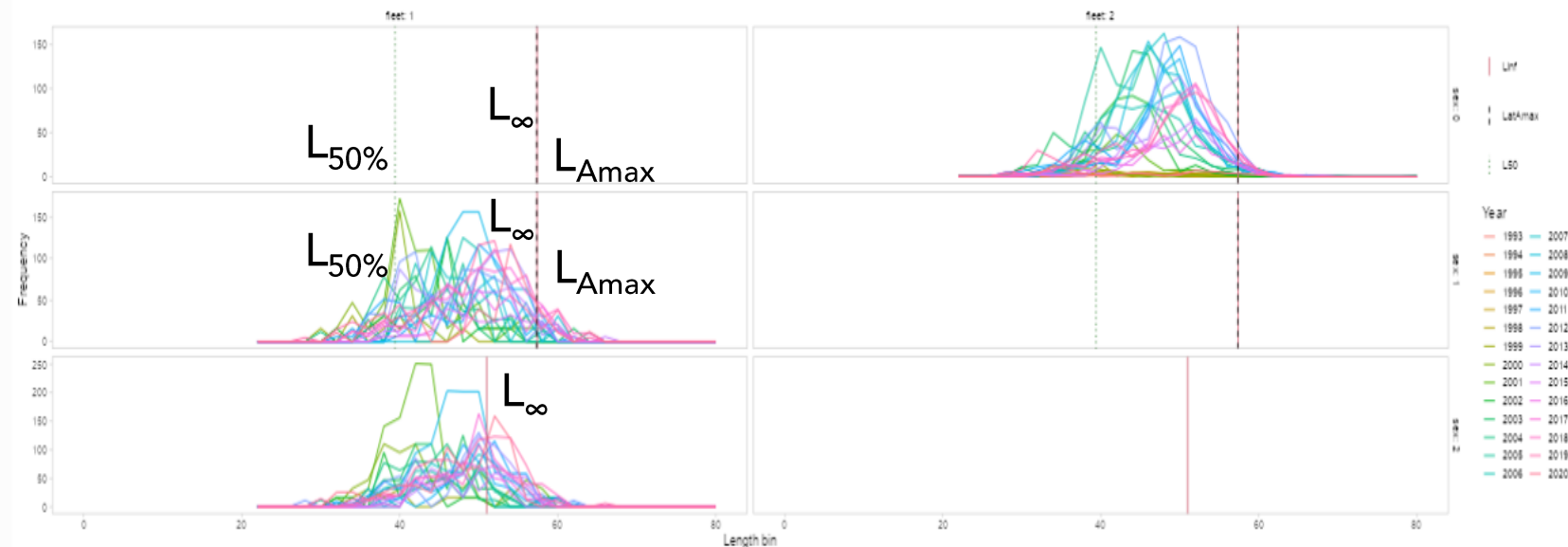
Enter male specific values. Otherwise, males are assumed equal to females

If estimating females and wanting to match males to estimated female parameters, use the male offset option and set offsets to 0. This will ensure male values are equal to estimated female males. Otherwise, males will retain the initial female values.

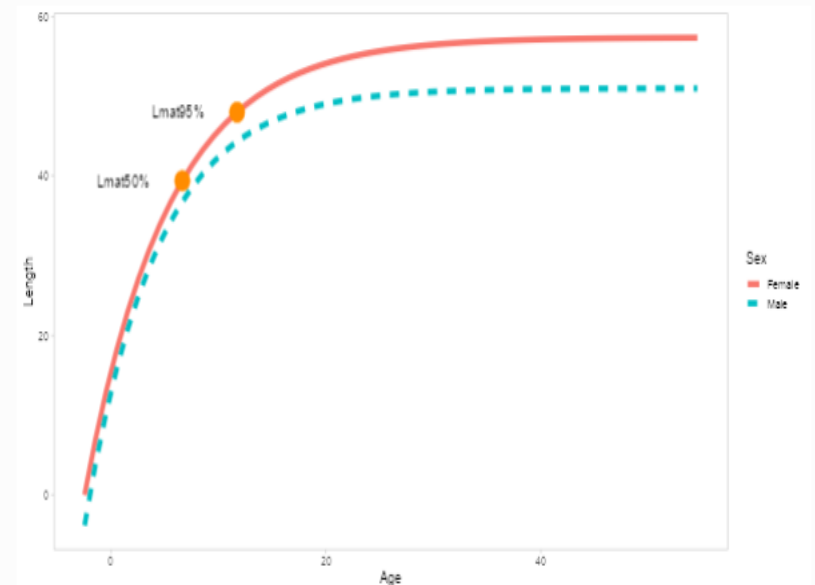
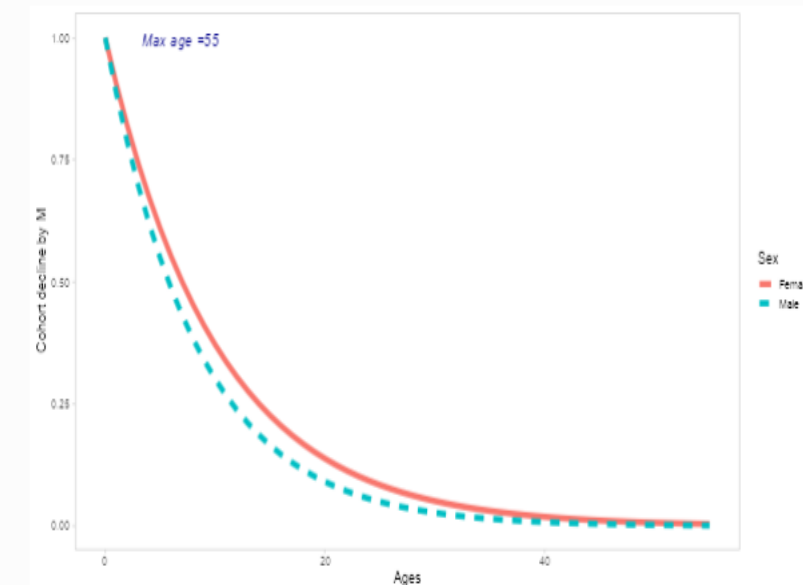
☒ Males specific values?

☐ Males offset from females

Length compositions. Sex categories: unknown sex = 0, females = 1 and males = 2.



Life history



SAC Tool: Recruitment

Stock-recruitment parameters

Stock-recruit type

Beverton-Holt

Steepness

0.72

Log initial recruitment

7

Recruitment month

1

Stock-recruitment function
S-R parameterization

☒ Estimate recruitment?

Recruitment estimation

Rec. variability (sR)

0.5

Recruit deviation option

3: deviation vector

Simple (option 2) or deviation vector (option 3)
recommended

Rec. devs. start year

1892

Rec. devs. end year

2020

☒ Bias correct recruitments?

Bias correction

Years of no bias correction

Early last year

1892

1st recent year

2020

Years of bias correction

Start year

1892

End year

2020

Maximum bias adjustment

1

- Choose years for no and full bias.
- Ramp between no and full bias
- Get these values using variance estimation on

SAC Tool: Selectivity

Selectivity

Enter parameter and phase values for each fleet and survey.

Example using 50% selectivity with two fleets: Inputs could be 35,40 and 2,2 for starting values and phases respectively.

The phase input indicates estimated parameters. To pre-specify the parameter, set the phase value to a negative number.

If using a mix of logistic and dome-shaped selectivities, select **dome-shaped** and pre-specify (i.e., use a negative phase) the provided default values (10000,0.0001,0.9999 for the additional parameters, respectively) to achieve a logistic shape for any given fleet.

Length selectivity type

Dome-shaped

Length at 50% Selectivity

35,40

Est. phase

3,3

Length at Peak Selectivity

40,45

Est. phase

3,3

Length at 1st declining selectivity

10000,48

Est. phase

-3,3

Width of declining selectivity

0.0001,10

Est. phase

-3,2

Selectivity at max bin size

0.99999,0

Est. phase

-3,2

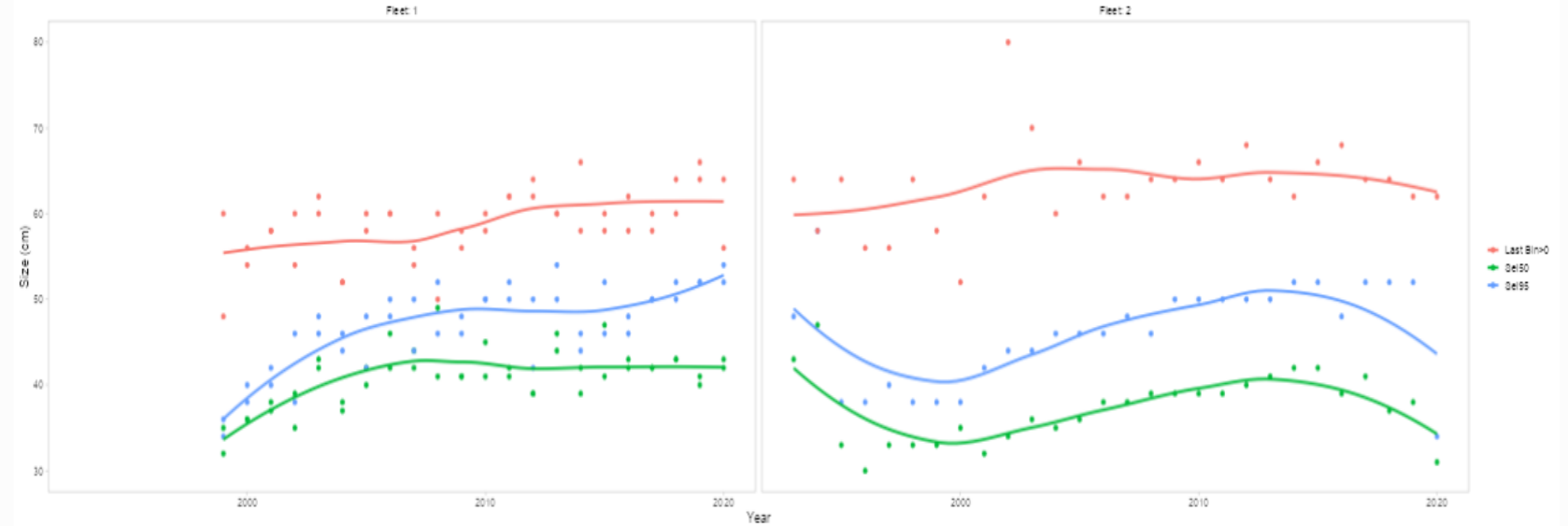
☐ Jitter starting values?

Jittering refers to changing the input starting values.

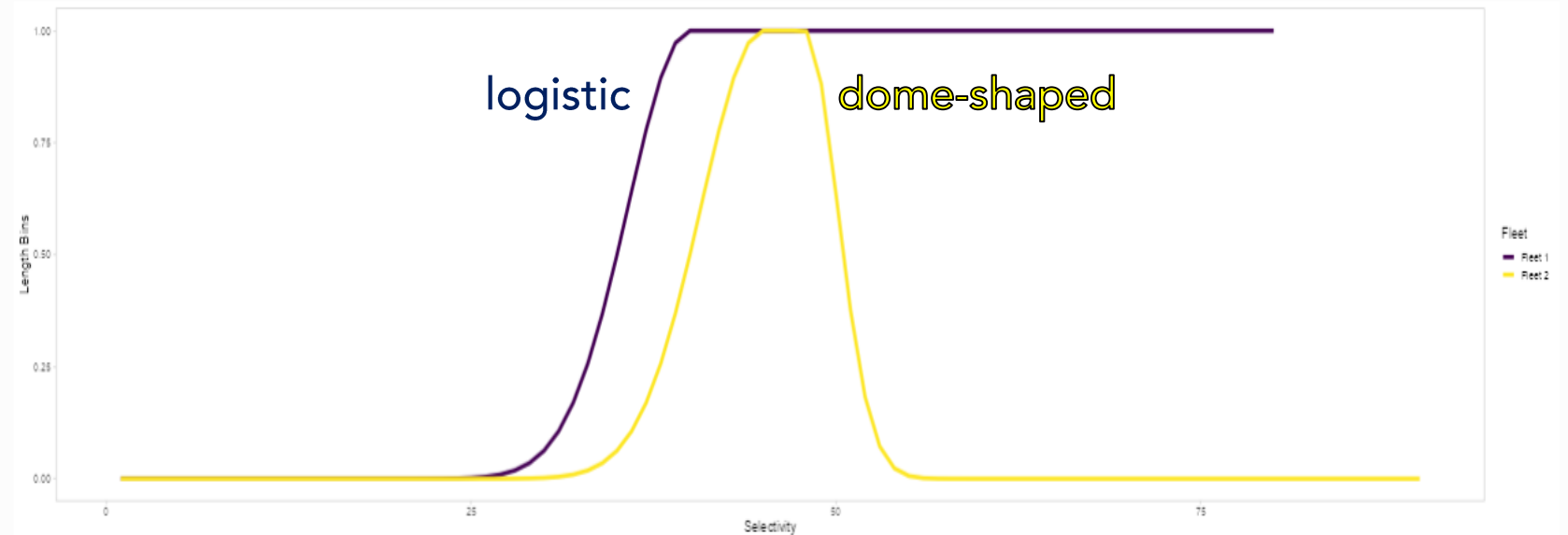
Jittering provides a quick way to adjust starting values for two main purposes:

- Start the model at different values to assist model convergence.

Suggested selectivity starting values based on length compositions by year. Lines are loess smoothers. Sel95 is based on the mode of the composition; Sel50 is based on the midpoint of the mode and first size of recorded lengths. Changes in values over time could be indicative of changes in selectivity, recruitment events, sampling representativeness, or other factors, and should be thought through when specifying selectivity for each fleet. These plots are informative to the behavior of the ascending limb of selectivity. More information is needed to specify the possibility of a descending limb (and thus form of the selectivity).



Selectivity



SAC Tool: Additional options

Additional SS3 options

Additional SS3 commands can be found [here](#)

☒ Turn off variance estimation (speeds up runs)

☐ Add additional SS run commands

☐ Turn off plots

Plot target and limit reference points

0.4,0.25

☒ No executive summary tables

☐ Turn off estimation of all parameters (phase = 0)?

☐ Specify non-biomass catch units for each fleet.

☐ Use only one growth type (default is 5)

☐ Estimate additional variance on each abundance index?

☐ Maintain sex ratio in biological compositions

☐ Add custom ageing error matrices?

Software to calculate ageing error matrices from multiple age reads is available [here](#).

Define modelled length bins

Default values are by 2 cm bin ranging from 4 to 25% above the Linf value. If using conditional age at lengths, length bins must be consistent with these population bins, not the length data bins.

Inputs must be smaller and larger than the length composition bins. Input values will be overridden to meet this requirement

bin size

2

minimum bin

4

maximum bin

72

Define plus group age

Default value is based on the female natural mortality value or the maximum age found in the age data file.

If you change this value here AND you are using age data, you will need to update the age data (and ageing error) to match the dimensions expressed here.

Plus group age

55

SAC Tool: Model output

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Scenario name

Choose the name of your scenario

Scenario_1

Each scenario folder is saved. Changing the scenario name therefore creates a new folder of results.

Using different scenario names when changing data or parameter values allows easy sensitivity exploration.

For Mac and Linux users, do not leave spaces in the Scenario name.

Save session inputs before model run

Worried about losing your inputs from a crashed model? Click the 'save session inputs' button BEFORE running the model to save your inputs.

To recovery those inputs, you can paste the link into another webpage. The SS-DL tool needs to be running to recovery another session. Note you will also need to reload data files to see the saved inputs.

You can also retain the link to share or report inputs. Consider adding a text file with the link into any model run folders for future explorations.

 Save session inputs

Data and Parameters

Model output

Model efficiency

Jitter exploration

Likelihood profile

Retrospectives

Sensitivity Plots

Ensemble models

Model output tab

Full model output is contained in the Report.sso file. The following reports are meant for quick examination.

Checking model convergence. Check also fit to length composition data

Maximum gradient: 4.03647e-05

Was covariance file created? FALSE

Model may not have converged or inputs are missing. Please use the Jitter option or check/change starting values before re-running model.

Key Derived Outputs

Search

Label ↕	Value ↕	StdDev ↕
SO2020/SO_0	0.28	0
SO_MSY/SO_0	0.29	0
1-SPR2020	0.74	0
OFL2021	4.06	0
ABC2021	3.08	0

1-5 of 5 rows Show 10 ▼

Previous 1 Next

Fishing Intensity

Search

SAC Tool: Model efficiency

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Model efficiency tab

Select model folder

Choose method to use

There are two main Bayesian methods to choose from:

- Random walk Metropolis (RWM). This method is useful to quickly explore parameter behavior.
- No u-turn (Nuts). This method can quickly run Bayesian models once an efficient model is established. No thinning necessary, as that is part of the algorithm

The following is a recommended work flow to find and run an efficient Bayesian model:

- Optimize your model with a short Bayesian run. Click the 'optimize model' button.
- Run your optimized model using RWM with 2000 iterations and thin = 10, then use the pairs plots to look for parameters that don't change value across kept draws.
- Evaluate the produced pairs plots to look for parameters that don't change value across kept draws.
- Parameters that don't move should be pre-specified in the model, and the model re-optimized.
- Run the RWM model again and continue looking for and pre-specifying any non-moving parameters.

Once the model specification is finalized, you can

- Re-run with Hessian (go back to the first tab to re-run model) to get asymptotic variance estimates.
- Re-optimize, then consider using the NUTS option with 1000 iterations or keep running the RWM option until convergence criteria are reached by increasing the thinning value and/or number of iterations. This produces a Bayesian representation of uncertainty.

Good model convergence is indicated when:

- Minimum effective sample size (ESS) > 200
- Rhat < 1.1. This measures the ratio of overestimated to underestimated variance.

☒ Optimize model?

One should include model optimization before running the evaluation methods below if this is the first run of a given model specification, including if parameters have been pre-specified since the last exploration.

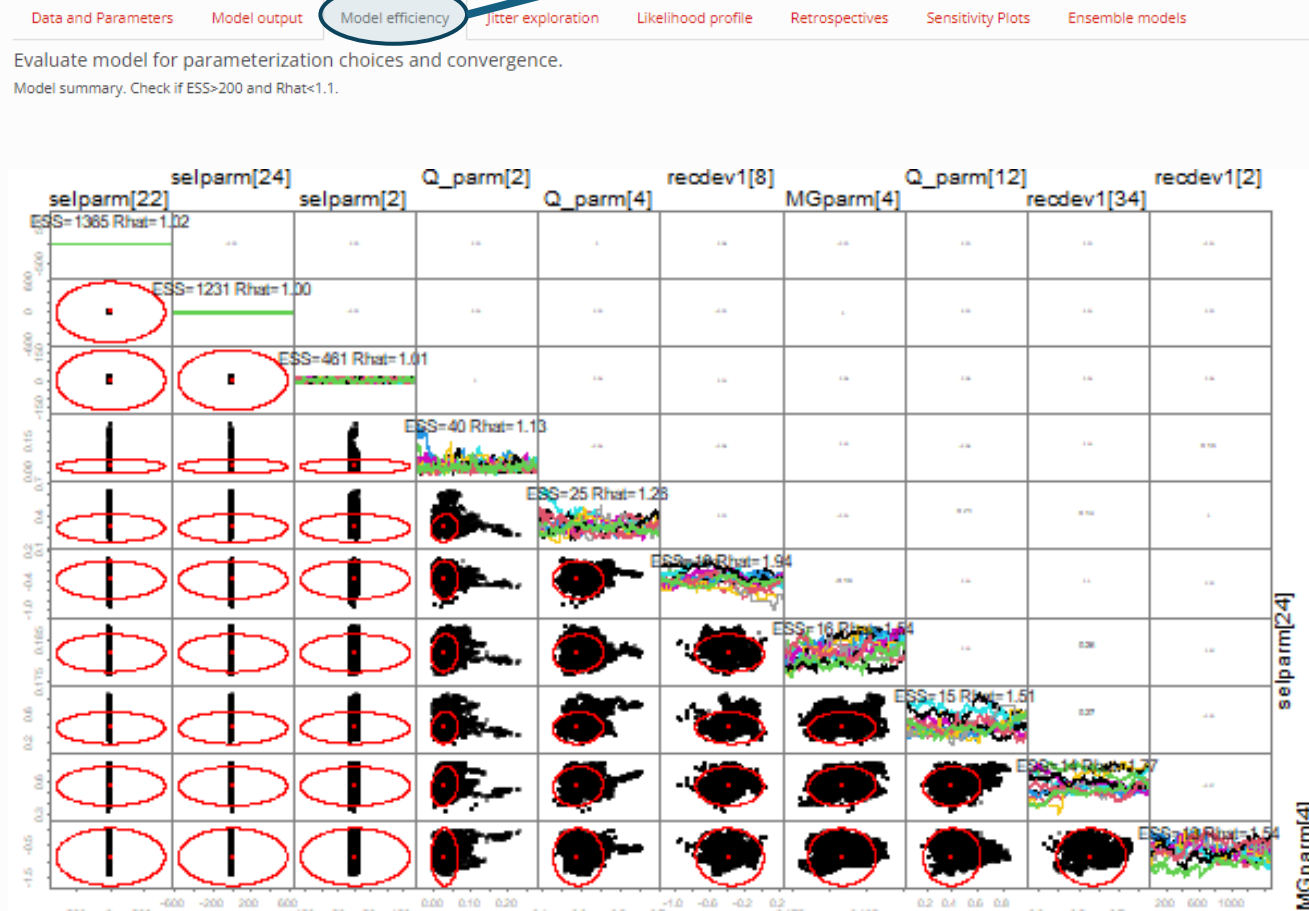
Choose method and evaluation inputs

☒ RWM ☐ Nuts

How many iterations to run?

Thinning (RWM only): # of iterations to keep?

0.1:3567/#tab-9323-4



SAC Tool: Jitter results

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Jitter tab

Data and Parameters

Model output

Model efficiency

Jitter exploration

Likelihood profile

Retrospectives

Sensitivity Plots

Ensemble models

Choose data file

Catch time series

Browse...

Length composition

Browse...

Age composition

Browse...

Abundance index

Browse...

Clear data files

Catches

Length

Ages

Index

☐ Use existing model files?

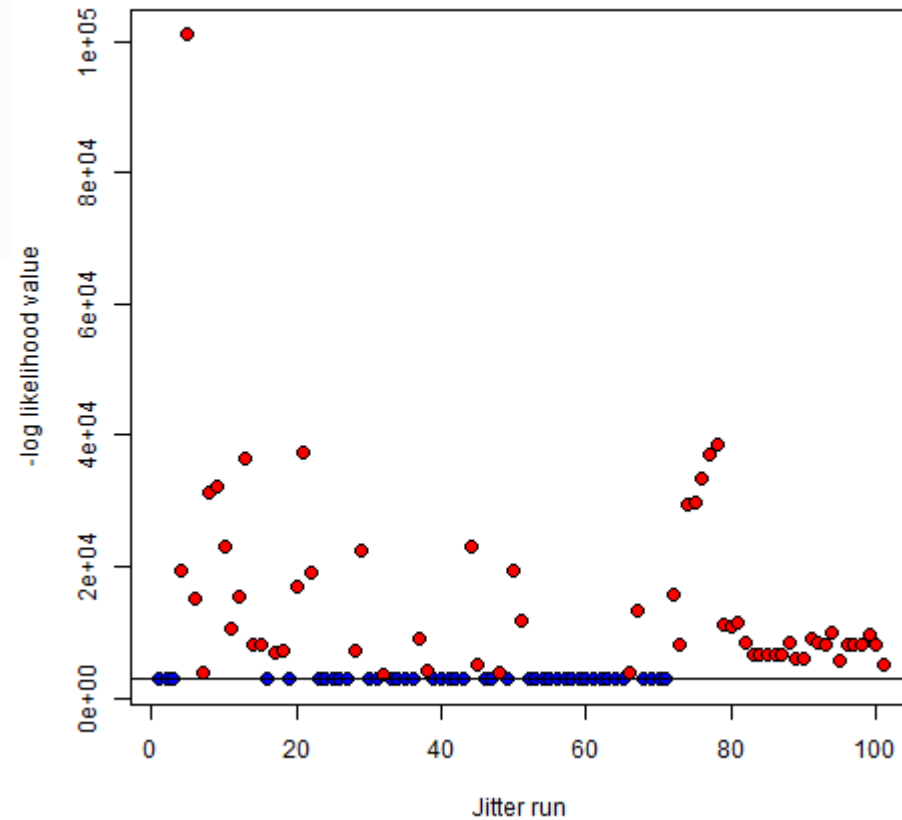
Do not use this option with catch only models.

Make sure the model run is in the Scenarios folder. Put that folder name in the Scenario name input and run the model.

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Examples are applying jitters or any of the additional SS3 options that do not modify the data or control files.

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SAC Tool: Likelihood profiles

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Likelihood profile

Run likelihood profiles

Likelihood profiles are a powerful way to understand the information content of data and sensitivity of models to parameter uncertainty.

A likelihood profile pre-specifies a chosen parameter to a specified set of values in a reference model. The reference model will maintain estimation of any other parameters estimated in the reference model.

For example, natural mortality (M) could be profiled over the value 0.1 to 0.3 at steps of 0.01. This creates 21 model runs that pre-specify M to different values while keeping all other specifications the same as the reference model.

For each model run, the likelihood value and derived outputs are retained for analysis.

Any likelihood values > 1.96 units from the minimum value are identify as models statistically less supported by the data.

Plots with the profiled parameter values compared to the likelihood values and derived model outputs indicate how much information is contained in the model for the parameter and how sensitive the model is to parameters values resulting non-statistically different models.

Choose folder of scenario to run profile

Select scenario

Choose model folder

Individual likelihood profiles- each parameter run independently.

If choosing multiple parameters to individually profile over, entries should be done in order of the parameters shown and separated by a comma (e.g., 0.1, 0.3).

The range of values must also include the value of the model being used. If not, the profile will not run.

Choose parameters to profile over

Nothing selected

Low value

High value

Sequence step

Run Likelihood Profile

Another option is to run simultaneous likelihood profiles.

For example, L_{inf} and k are negatively correlated and should be changing together in a profile. Read in .csv file with vectors of values for each parameter.

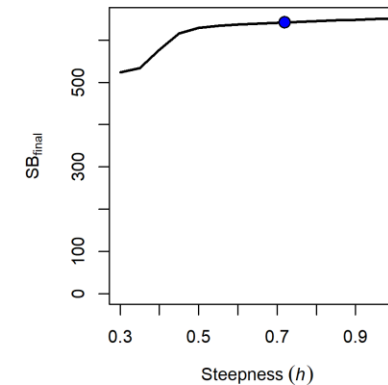
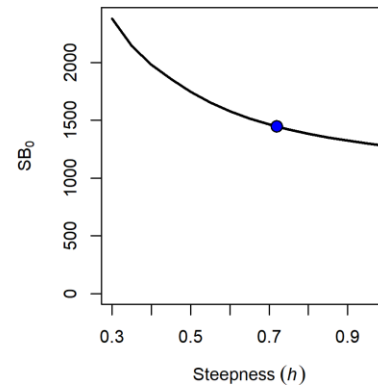
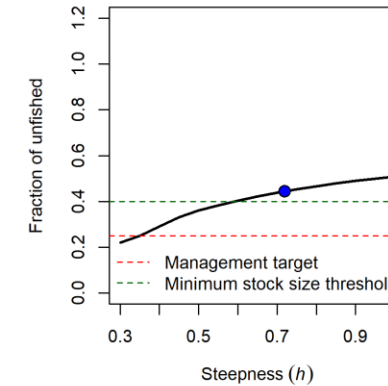
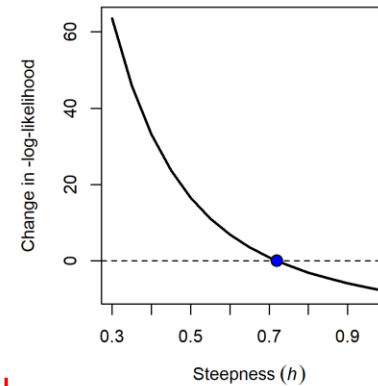
See the 'Multi_profile_headers.csv' file for parameter header names. Erase columns not used

Data and Parameters Model output Model efficiency Jitter exploration Likelihood profile Retrospectives Sensitivity Plots Ensemble models

Full likelihood profile outputs and plots can be found in the 'Profile' folder of the chosen scenario.

Each profile folder will be labeled with the parameter name appended.

Results below are from the one of the profile parameters to show what type of plots are found in the folders.



Choose parameter(s) to profile and values

SAC Tool: Retrospective plots

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Retrospectives

Data and Parameters Model output Model efficiency Jitter exploration Likelihood profile Retrospectives Sensitivity Plots Ensemble models

Retrospective comparisons and plots

Retrospective modelling means sequentially removing one year of data up to a specified number of years (e.g., -10 years).

To make these comparisons, choose first the directory containing models, then the models to compare.

A time series plot of comparisons are shown in the main panel to the right for the following model outputs:

- Spawning output
- Relative spawning output
- Recruitment

A complete complement of comparison plots (along with the plot on the right) are saved in the chosen folder labeled 'retro'

Choose folder containing model for retrospective analysis

Select folder

Choose model folder

Define what years to perform retrospective analysis. Input as a negative integer (e.g., -1 mean remove one year of data)

1st retrospective year

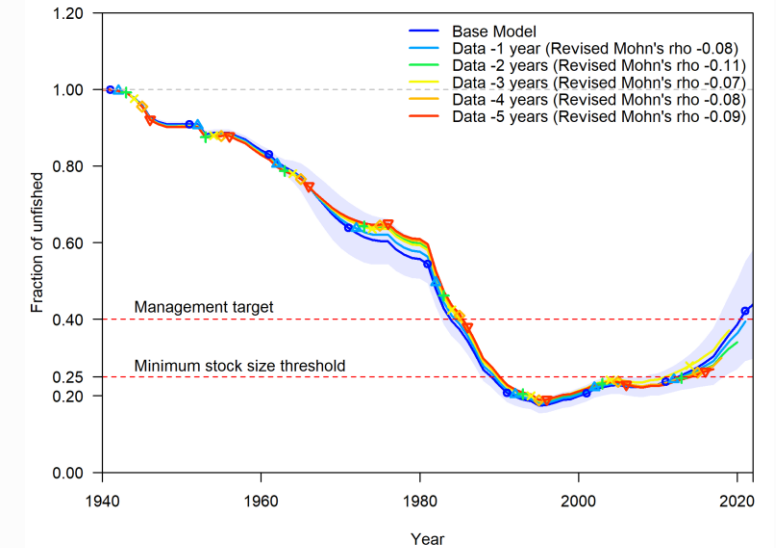
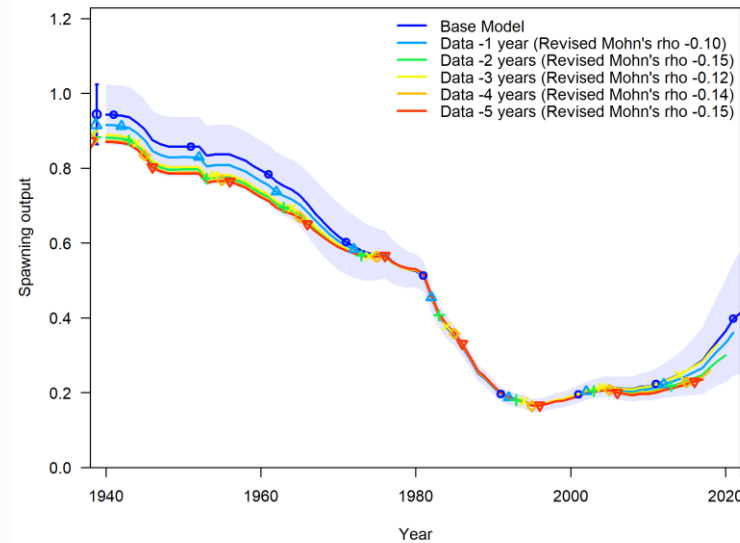
-1

Last retrospective year

-5

Choose years

Run Retrospective Comparisons



SAC Tool: Sensitivity plots

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Sensitivity Plots

Sensitivity comparison plots
Comparing models offers insight into how changing data or model specification change model outputs.
To make these comparisons, choose first the directory containing models, then the models to compare.
A time series plot of comparisons are shown in the main panel to the right for the following model outputs:

- Spawning output
- Relative spawning output
- Recruitment

A complete compliment of comparison plots (along with the plot on the right) are saved in the chosen directory in a folder labeled

Choose folder containing model scenarios

Select directory

Choose reference model

Nothing selected

Choose scenarios to compare to reference model

Nothing selected

Add reference points to spawning output plots. Blank input adds no line.

Target

Limit

Sensitivity relative error plot features

Add y-axis limits, vertical subheader breaks, sensitivity group subheaders, text size and positioning

Minimum y-axis value

Maximum y-axis value

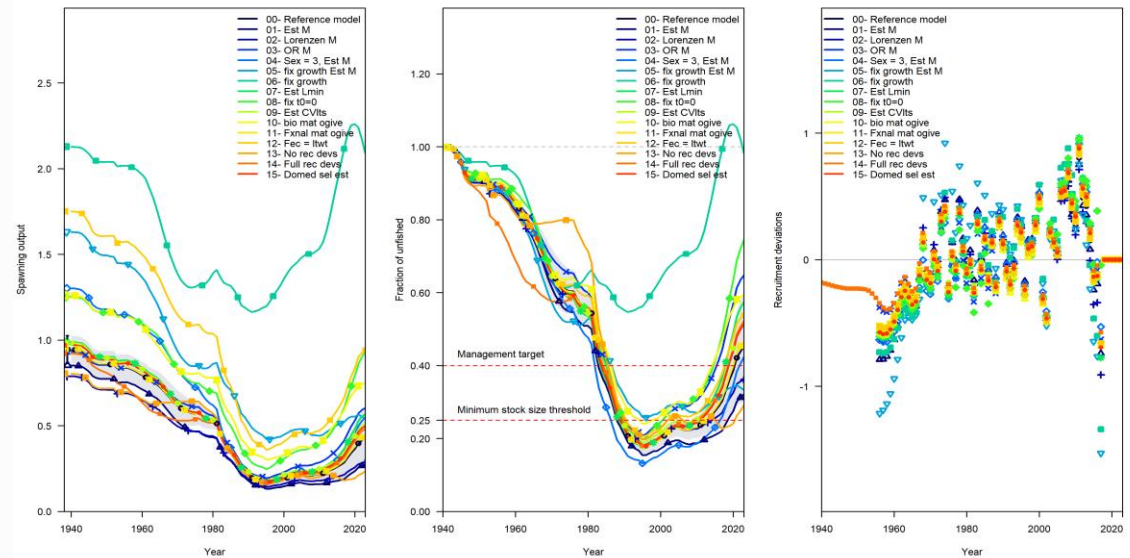
-1

1

Subheader vertical break positions

Data and Parameters Model output Model efficiency Jitter exploration Likelihood profile Retrospectives **Sensitivity Plots** Ensemble models

Time series sensitivity plots



Relative change (Scenario relative to reference model) sensitivity plots for 5 different metrics

Unfished spawning output and terminal year spawning output measure scale

Relative biomass is a stock status measure

FMSY is a measure of productivity; MSY is a measure of productivity and scale

For more details on interpreting these plots, please see [Cope and Gertseva 2020](#)

SAC Tool: Ensemble models

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Ensemble
models

Ensemble modelling

Ensemble modelling allows the user to combine multiple models into one weighted distribution of outputs.

User first chooses the models to combine, then how to combine them via model weights.

For example, if 3 models are chosen, weights of 1,1,1 defines equal weights.

If the middle model should have twice the weight of the others, 1,2,1 is the weighting input.

Choose folder containing models to combine

Select directory

Choose directory

Label ensemble model file

Ensemble 1

Choose scenarios to ensemble

Nothing selected

Choose models to combine

Relative scenario weights

Choose model weights

Create Ensemble Model

